

Education

Boston University (BU)

Ph.D., Computer Science

Advisor: [Edward Chien](#)

Research Focus: Computer Graphics, Geometry Processing, Digital Fabrication

Sept 2021 - present

Trinity College (TC), CT

B.Sc. Physics (honors), B.Sc. Computer Science (honors)

Summa Cum Laude, Phi Beta Kappa, Sigma Pi Sigma (Physics honor society) inductee

Advisor: [Kevin Huang](#)

Research Focus: Telerobotics, Haptic User Interfaces, Contact Sensing

Sept 2017 - May 2021

GPA: 3.95/4.00

Technical skills

Programming Software/Libraries

C++/C, Python, Java, Mathematica, MATLAB, OpenGL Shading Language (GLSL)
Blender, MeshLab (3D modelling), Gurobi (optimization), LibIGL, geometry-central
(geometric algorithms), polyscope (3D visualization), Eigen (numerical linear algebra), OpenGL (graphics programming), Git

Work & Research Experience

Graduate Research Assistant, Computer Graphics Lab

Boston University. Advisor: [Edward Chien](#)

Sept 2021 - present

Boston, USA

- Conducting research in applying geometry processing techniques for tasks in digital fabrication, vector field design and remeshing.
- *Stripe Patterns for Computational Knitting*: Designed and implemented a striping algorithm for generating machine-knittable graphs over input models. Our approach introduces several key techniques that allows for user-design choices while ensuring machine-knittability. See Pub. [3].
- *Foliations for Computational Knitting*: Enhanced the method of Pub. [3] by abstracting stripe patterns as foliations (integral curve lines) of a vector field. This perspective allows for more robust graph generation, further expanding the user-design space and diversity of input models. See Pub. [2].
- *Automatic Knit Singularity Placement*: Designed and implemented a greedy and efficient knit singularity placement algorithm. This allows for the realistic yarn-level rendering and fabrication of 2D & 3D models with high geometric fidelity. See Pub. [1].

Game Development and Research Intern, Lightspeed Studios × 2

Tencent Americas. Mentor: [Kui Wu](#)

Summer 2024, Fall 2024

Los Angeles, USA

- *Quad-dominant meshing for rendering/knitting*: Implemented a pipeline that uses custom striping algorithms to generate quad-dominant elements over cut & sew patterns (and extends to general 3D models) with the goal of developing novel meshing techniques for rendering and knitting of yarn-level models. See Pub. [1].

Researcher, High Performance Computing Group

Trinity College. Advisor: [Peter Yoon](#)

Sept 2020 - May 2021

Hartford, USA

- Implemented a system to provide real-time audio feedback on a white cane using NVIDIA's Jetson Nano microcomputer. Integrated the Raspberry Pi V2 Camera with the Nano for real-time video input. Used the ssd-inception-v2 model and tensorflow for image classification. Developed application for audio feedback and seamless bluetooth interfacing between end-user and Nano.
- Winner of best thesis project based on completeness, technical maturity and relevance.

Researcher, Perceptual Robotics & Automation Lab

Trinity College. Advisor: Kevin Huang

Sept 2018 - May 2021

Hartford, USA

- *Vibration-based sensor*: Modelled contact-sensing as vibration-classification problem. Classified data using Gaussian mixture model clustering and logistic regression. Built system to interface sensor with Raspberry Pi microcomputer (used for data collection). Pub. [5].
- *Vision-based force-feedback in Robot-Assisted surgery*: Examined deviation of haptic feedback from ground truth for acceptable performance in Robot-Assisted Surgery. Explored models for node-to-node interaction in simulated tissue surface. Pub. [6].
- *Haptic Interface for Robot Locomotion*: Developed software and experimental protocol to compare a haptic interface vs keyboard and joystick interface for legged robot-locomotion. Conducted user-studies and statistical analysis of results. Pub. [4] & Pub. [7].
- *Joint-limit haptic feedback*: Implemented point cloud generation/retrieval models for providing haptic feedback in teleoperated robots. Presented paper at ICARM. Pub. [8].

Publications (chronological)

- [1] **Mitra, R.**, Couplet M., Wang, T., Hofmann, M., Wu, K., & Chien, E. (2025, July). Curl Quantization for Automatic Placement of Knit Singularities. In ACM SIGGRAPH 2025 Conference Proceedings (pp. 1-10).
- [2] **Mitra, R.**, Jimenez Berumen, E., Hofmann, M., & Chien, E. (2024, July). Singular Foliations for Knit Graph Design. In ACM SIGGRAPH 2024 Conference Papers (pp. 1-11).
- [3] **Mitra, R.**, Makatura, L., Whiting, E., & Chien, E. (2023, July). Helix-Free Stripes for Knit Graph Design. In ACM SIGGRAPH 2023 Conference Proceedings (pp. 1-9).
- [4] Huang, K., Subedi, D., **Mitra, R.**, Yung, I., Boyd, K., Aldrich, E., & Chitrakar, D. (2020). Telelocomotion—remotely operated legged robots. *Applied Sciences*, 11(1), 194.
- [5] **Mitra, R.**, Boyd, K., Subedi, D., Chitrakar, D., Aldrich, E., Swamy, A., & Huang, K. (2020, October). Contact sensing via active oscillatory actuation. In 2020 3rd International Conference on Mechatronics, Robotics and Automation (ICMRA) (pp. 99-104). IEEE.
- [6] Huang, K., Chitrakar, D., **Mitra, R.**, Subedi, D., & Su, Y. H. (2020, July). Characterizing limits of vision-based force feedback in simulated surgical tool-tissue interaction. In 2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) (pp. 4903-4908). IEEE.
- [7] Chitrakar, D., **Mitra, R.**, & Huang, K. (2020, November). Haptic interface for hexapod gait execution. In 2020 Fourth IEEE International Conference on Robotic Computing (IRC) (pp. 414-415). IEEE.
- [8] Huang, K., Su, Y. H., Khalil, M., Melesse, D., & **Mitra, R.** (2019, July). Sampling of 3dof robot manipulator joint-limits for haptic feedback. In 2019 IEEE 4th International Conference on Advanced Robotics and Mechatronics (ICARM) (pp. 690-696). IEEE.

Teaching & Service

Instructor

Intro. to CS, BU Summer Challenge. Summer '25.

Reviewer

UIST, '25, SIGCHI, '25.

Teaching Assistant, Geometric Algorithms × 2

Fall '23, Fall '25, BU.

Mentor, Summer Geometry Initiative (SGI)

Teaching Assistant, Geometry Processing (Graduate Course) × 2

Teaching Assistant, Data Structures & Algorithms × 2

Teaching Assistant, Classical Mechanics

Teaching Assistant, Introduction to Computing

Teaching Assistant, Mobile Robotics

Robotics mentor, Tech Savvy

Program to introduce geometry processing research to students globally. Summer '23, MIT.

Spring '23, Spring '24, BU.

Spring '20, Spring '21, TC.

Fall '20, TC.

Spring '19, TC.

Spring '19, TC.

American Association of University Women (AAUW) program introducing careers in STEM to middle school girls. Spring '18, '19.

Introductory robotics program for Hartford middle schools.

Volunteer Teacher, Hartford Teach the Teachers

Talks/Presentations

“Power Diagrams on Surfaces

for Placement of Knit Singularities”

Symp. on Computational Fabrication Nov, '25

“Curl Quantization for Automatic Placement

of Knit Singularities” (Poster)

New England Symp. on Graphics Apr, '25

“Computational Knitting”

CS680: Computer Graphics, BU Dec, '24

“Singular Foliations for Knit Graph Design”

SIGGRAPH Jul, '24

“Singular Foliations for Knit Graph Design” (Poster)

Eurographics SGP Jun, '24

“Helix-Free Stripes for Knit Graph Design”

SIGGRAPH Aug, '23

“Stripe Patterns for Computational Knitting”

BU Graphics Seminar '22 - '24

“Sampling of 3DOF Robot Manipulator

Joint-limits for Haptic Feedback”

IEEE ICARM Jul, '19

Awards

Eurographics Widening Participation Scholarship

Eurographics. 2024

Best Computer Science Senior Thesis

Trinity College. 2021

President’s Fellow in Physics, Class of 2021

Awarded to the strongest major in the graduating

class. Trinity College. 2021

Albert J. Howard Jr. Prize in Physics

Awarded to the strongest major in the junior class.

Trintiy College. 2020

Phi Gamma Delta Prize in Mathematics

Awarded for outstanding performance in mathe-

matics coursework. Trinity College. 2020

Faculty Honors

Trinity College. All Semesters

Full Tuition Scholarship

Trintiy College. 2017